
Brief Explanation of Transformation and Modeling Logic

Step 1: Data Import

The datasets List of Orders, Order Details, and Sales Target were imported into Power BI using the Get Data option. The Transform Data option was used to open Power Query Editor for data preparation and transformation.

Tables Imported

- List of Orders
 - Order Details
 - Sales Target
-

Step 2: Row Limiting

The List of Orders table contained more than 500 records. As per the assignment requirement, only the first 500 rows were retained.

Power BI Steps

Home → Keep Rows → Keep Top Rows → Enter 500

Purpose

To work with a controlled dataset while performing data transformation and analysis.

Step 3: Data Type Conversion

The data types were reviewed and corrected to ensure accurate calculations and reporting.

Changes Made

Order Date

Transform → Data Type → Date

Amount

Transform → Data Type → Fixed Decimal Number

Target

Transform → Data Type → Fixed Decimal Number

Purpose

Correct data types improve calculation accuracy and data model performance.

Step 4: Text Formatting

The CustomerName column was converted into Proper Case format to maintain consistent capitalization.

Power BI Steps

Select CustomerName

Transform → Format → Capitalize Each Word

Example

Before: bharat kumar

After: Bharat Kumar

Purpose

To improve data consistency and readability.

Step 5: Creating Location Column

The City and State columns were merged to create a new column named Location.

Formula

Location = [City] & ", " & [State]

Example

Chennai, Tamil Nadu

Mumbai, Maharashtra

Power BI Steps

Select City and State columns

Transform → Merge Columns

Separator = Comma

New Column Name = Location

Purpose

To simplify geographical analysis.

Step 6: Creating Profit Margin

After merging the datasets, a custom column named Profit Margin was created.

Formula

Profit Margin = Profit / Amount

Power BI Steps

Add Column → Custom Column

Formula:

[Profit] / [Amount]

The resulting values were formatted as percentages.

Purpose

To measure profitability relative to sales amount.

Step 7: Creating Profit Status

A conditional column named Profit Status was created based on Profit values.

Business Logic

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

Formula

```
if [Profit] < 0 then "Loss"  
else if [Profit] = 0 then "Break-Even"  
else "Profit"
```

Purpose

To classify orders based on profit performance.

Step 8: Merging Tables

The List of Orders and Order Details tables were merged using Order ID.

Power BI Steps

Home → Merge Queries as New

First Table = List of Orders

Second Table = Order Details

Join Column = Order ID

Join Type = Left Outer Join

Output Table

Orders Data

Purpose

To create a consolidated dataset containing both order-level and product-level information.

Step 9: Handling Missing Values

A data quality check was performed using the Column Quality feature in Power Query Editor.

Power BI Steps

View → Column Quality

Findings

After restricting the List of Orders table to the first 500 rows, no missing values were found in the final dataset used for analysis.

Action Taken

No replacement or removal of records was required.

Justification

Since the dataset contained complete records, all rows were retained to preserve maximum information for analysis.

Step 10: Handling Duplicate Records

Duplicate records were checked in all tables.

Power BI Steps

Home → Remove Rows → Remove Duplicates

Findings

No duplicate records were found in the final dataset used for analysis.

Action Taken

No duplicate removal was required.

Justification

The dataset contained unique records, ensuring accurate aggregation and reporting.

Step 11: Sorting Data

The Orders Data table was sorted by Order Date in descending order.

Power BI Steps

Select Order Date

Sort Descending

Purpose

To display the latest transactions first.

Step 12: Filtering Data

A filter was applied to analyze orders from Tamil Nadu.

Power BI Steps

Filter Pane → State → Tamil Nadu

Purpose

To perform state-level sales analysis.

Step 13: Aggregation – Count of Orders

The Order Details table was duplicated and grouped to calculate the total number of orders.

Power BI Steps

Right Click Order Details → Duplicate

Home → Group By

Operation = Count Rows

New Column Name = Order Count

Purpose

To determine the total order volume.

Step 14: Aggregation – Average Profit by Category

A duplicate of the Order Details table was grouped by Category.

Power BI Steps

Group By → Category

Operation = Average

Column = Profit

Purpose

To compare profitability across product categories.

Step 15: Aggregation – Total Amount by Sub-Category

A duplicate of the Order Details table was grouped by Sub-Category.

Power BI Steps

Group By → Sub-Category

Operation = Sum

Column = Amount

Purpose

To identify high-revenue product groups.

Step 16: Aggregation – Monthly Sales Target

The Sales Target table was duplicated and grouped by Month of Order Date.

Power BI Steps

Right Click Sales Target → Duplicate

Home → Group By

Group By = Month of Order Date

Operation = Sum

Column = Target

Purpose

To calculate total monthly sales targets.

Step 17: Data Modeling

Relationships were established in Model View.

Relationship 1

List of Orders[Order ID]

↓

Order Details[Order ID]

Cardinality: One-to-Many

Status: Active

Relationship 2

Order Details[Category]

↓

Step 18: Relationship Validation

Sales Target[Category]

Manage Relationships was used to verify that all relationships were active and correctly configured.

Cardinality: Many-to-Many

Checks Performed

- Status: Active
- Correct join columns
 - Active relationships

Purpose

To ensure proper cardinality

Purpose

To enable data integration and cross-table

analysis. To ensure accurate filtering, aggregation, and reporting throughout the data model.

Conclusion

The project involved importing data, performing transformations, correcting data types, creating calculated and conditional columns, merging datasets, validating data quality, performing aggregations, and establishing relationships between tables. These steps produced a clean, integrated, and analysis-ready data model suitable for business reporting and decision-making in Power BI.

Screenshots:

Table view

Order Details

- Category
- Order ID
- Σ Amount
- Σ Profit
- Σ Profit Margin
- Profit Status
- Collapse ^

List of Orders

- Order Date
- Order ID
- CustomerName
- Location
- Collapse ^

Orders Data

- Order Date
- CustomerName
- Location
- Order ID
- Collapse ^

Average Profit

- Σ Average Profit
- Category
- Collapse ^

Data

Tables**Model**

- > Average Profit by category
- > Count of orders
- > List of Orders
- > Order Details
- > Orders Data
- > Sales target
- > Sales target By Month
- > Total by subcategory

Queries [8]

Sales target
= Table.TransformColumnTypes(#"Promoted Headers",({{"Month of Order Date", type date}, {"Category", type text}}))

	Month of Order Date	Category	Target
1	01-04-2018	Furniture	20,400.00
2	01-05-2018	Furniture	20,500.00
3	01-06-2018	Furniture	20,600.00
4	01-07-2018	Furniture	20,800.00
5	01-08-2018	Furniture	20,900.00
6	01-09-2018	Furniture	21,000.00
7	01-10-2018	Furniture	21,100.00
8	01-11-2018	Furniture	21,300.00
9	01-12-2018	Furniture	21,400.00
10	01-01-2019	Furniture	21,500.00
11	01-02-2019	Furniture	21,600.00
12	01-03-2019	Furniture	21,800.00
13	01-04-2018	Clothing	22,000.00
14	01-05-2018	Clothing	22,000.00
15	01-06-2018	Clothing	22,000.00

Query Settings

- PROPERTIES
 - Name: Sales target
 - Sales target
 - All Properties
- APPLIED STEPS
 - Source
 - Promoted Headers
 - Changed Type

Queries [8]

Order Details
= Table.Distinct(#"Renamed Columns", {"Order ID"})

	Order ID	Amount	Profit	Quantity	Category
1	B-25601	1,275.00	-2148	7	Furniture
2	B-25602	168.00	-111	2	Electronics
3	B-25603	1,355.00	-60	3	Clothing
4	B-25604	65.00	17	2	Clothing
5	B-25605	75.00	0	7	Clothing
6	B-25606	87.00	4	2	Clothing
7	B-25607	50.00	25	4	Clothing
8	B-25608	1,364.00	-1864	5	Furniture
9	B-25609	485.00	29	4	Electronics
10	B-25610	1,076.00	-38	4	Electronics
11	B-25611	180.00	-59	2	Clothing
12	B-25612	259.00	-55	2	Furniture
13	B-25613	1,603.00	0	9	Clothing
14	B-25614	494.00	54	4	Furniture
15	B-25615	68.00	20	5	Clothing
16	B-25616	42.00	12	5	Clothing
17	B-25617	305.00	-270	5	Electronics

Query Settings

- PROPERTIES
 - Name: Order Details
 - Order Details
 - All Properties
- APPLIED STEPS
 - Source
 - Promoted Headers
 - Changed Type
 - Added Custom
 - Renamed Columns
 - Changed Type to percentage
 - Added Custom1
 - Renamed Columns1
 - Removed Duplicates

Queries [8]

List of Orders
= Table.Distinct(#"Merged Columns", {"Order ID"})

	Order ID	Order Date	CustomerName	Location
1	B-25601	01-04-2018	Bharat	Ahmedabad,Gujarat
2	B-25602	01-04-2018	Pearl	Pune,Maharashtra
3	B-25603	03-04-2018	Jahan	Bhopal,Madhya Pradesh
4	B-25604	03-04-2018	Divsha	Jaipur,Rajasthan
5	B-25605	05-04-2018	Kasheen	Kolkata,West Bengal
6	B-25606	06-04-2018	Izzet	Bangalore,Karnataka
7	B-25607	06-04-2018	Sonakshi	Kashmir,Jammu and Kashmir
8	B-25608	08-04-2018	Aasruthi	Chennai,Tamil Nadu
9	B-25609	09-04-2018	Ittezh	Lucknow,Uttar Pradesh
10	B-25610	09-04-2018	Yogesh	Patna,Bihar
11	B-25611	11-04-2018	Anita	Thiruvananthapuram,Kerala
12	B-25612	12-04-2018	Shrichand	Chandigarh,Punjab
13	B-25613	12-04-2018	Mukesh	Chandigarh,Haryana
14	B-25614	13-04-2018	Vandana	Simla,Himachal Pradesh
15	B-25615	15-04-2018	Bhavana	Gangtok,Sikkim
16	B-25616	15-04-2018	Kanak	Goo,Goo
17	B-25617	17-04-2018	Sagar	Kohima,Nagaland

Query Settings

- PROPERTIES
 - Name: List of Orders
 - List of Orders
 - All Properties
- APPLIED STEPS
 - Source
 - Promoted Headers
 - Changed Type
 - Kept 1st 500 rows of list of or...
 - Changed date type to date
 - Cleaned Text
 - Trimmed Text
 - Capitalized Each Word
 - Merged Columns
 - Removed Duplicates